Generating shot events (SH) from tracking data

Member 4: Shooting, Goalkeeping & Deflections · Mohamed Hatem · 11 Nov 2026 · Data: PFF FC World Cup 2022 (64 matches)

Result in one line: from tracking alone we find **60% of PFF's shots**, and **69% of the shots we report are real** (F1 0.64), on 16 matches the model never saw. For the shots we find, the shooter is right 70% of the time and the outcome 66%.

1. Task and data

PFF's event file is written by people watching video. The goal is to produce the shot event and its fields automatically, using only the tracking file (ball and player positions, 29.97 frames per second), the roster and the metadata. PFF's events are used only as the answer key for training and scoring.

Set	Matches	PFF shots	Use
Training	48	1,155	Build the model. Scored by 6-fold cross-validation with whole matches held out, so each match is predicted by a model that never saw it
Unseen	16	404	Fixed by random seed before any result was seen; used only for the final test

2. How we get the event, step by step

Step	What we do	Number
1. Read safely	Use raw ball and player positions only. PFF's event fields inside the tracking file and PFF's "smoothed" positions are deleted, because both contain the answer	Leakage test passes
2. Clean the ball	Remove jumps over 1.5 m, fill holes up to 0.2 s, smooth (Savitzky-Golay), compute velocity	Ball tracked in 62–78% of frames
3. Find every touch	A touch is a sudden change of the ball's velocity: average velocity of the 5 frames after minus the 5 frames before is at least 3 m/s	164,624 touches; 1,106 of 1,155 shots have one within 0.5 s
4. Physical filter	Keep touches within 36 m of a goal whose ball gets within 22.5 m of the goal line in 2 s	51,034 touches kept; 8 shots lost
5. Shot or not	About 115 numbers per touch (distance and angle to goal, ball speed and path to the goal mouth, keeper and defenders near the path, next touch, ball and player positions from 0.5 s before to 1.2 s after). Average of Gradient Boosting and Random Forest gives a probability; above 0.22 is a shot; one shot per 1.5 s	1 shot per 142 touches, so we score with precision, recall and F1
6. Shooter	Every attacking player is a candidate, described by his distance to the ball around the kick; a model picks the most likely one. The attacking team comes from the nearest goal and the metadata	
7. Outcome	Follow the ball for 3 s: who touches it next, whether and where it crosses the goal line, and how play restarts (players back in their own halves and long dead time indicate a goal). Gradient Boosting predicts PFF's 7 types	74% of real goals labelled as goals
8. Other fields	Keeper: defending goalkeeper on the pitch, written for saves and goals as PFF does. Blocker and failed blocker: candidate ranking over defenders near the ball's path. Heights, shot nature, ball moving, rebound zone: small models on ball height, speed and position	

3. Accuracy

Shot detection	Training CV	Unseen
Shots reported	1,004	352
Correct (within 0.5 s)	704	243
Precision (reported shots that are real)	70%	**69%**
Recall (PFF shots found)	61%	**60%**
F1 (combines both)	0.65	**0.64**
F1 per match: lowest / median / highest	–	0.36 / 0.65 / 0.87

Field (on detected shots)	Training CV	Unseen	
Ball moving	99%	99%	guess = 93%
Shooter	79%	70%	good
Outcome, 4 groups	68%	68%	good
Outcome, 7 PFF types	66%	66%	good
Keeper	68%	66%	good
Shot nature	71%	68%	guess = 49%

Timing error (median)	3 frames	3 frames

Reference: the best published tracking-only method (Vidal-Codina et al. 2022) reports precision 78% and recall 53% for shots, on different data.

Threshold (training CV)	Precision	Recall
0.15 (find more)	62%	67%
0.22 (used)	70%	61%
0.30	77%	56%
0.50 (rarely wrong)	88%	41%

Shooter movement	67%	64%	**no better than guess**
Shot height	49%	49%	**guess = 39%**
Failed blocker	43%	36%	**weak**
Save height	28%	33%	**no better than guess**
Blocker	20%	31%	**weak**
Save rebound zone	14%	12%	**weak**
Goal-line clearer	0%	0%	**8 examples only**

Shared part: naming who blocked a pass / cross / clearance, given the event moment and team (training CV): 69% / 76% / 48%.

4. Evidence that the model is sound

- **No overfitting in the reported numbers:** training CV and unseen agree (F1 0.65 vs 0.64; outcome 66% vs 66%). Only the shooter drops (79% to 70%).
- **No leakage:** deleting or scrambling PFF's event fields in the tracking file leaves the model input identical.
- **Better than rules:** hand-written rules alone reached F1 0.34 (measured on the first 6 matches).
- **Comparable to published work** (precision 78%, recall 53%), with tracking that PFF derives from TV broadcast.

5. Obstacles and what we did

Obstacle	Evidence	Action and status
Answer hidden in the tracking file	PFF's smoothed ball sits exactly on the shooter at shot frames	Use raw positions only; leakage test. **Solved**
Shots are rare	1 per 142 touches	Physical filter, then model; F1 instead of accuracy. **Solved**
Goals hard to recognise	Ball disappears in the net; replays hide the kick-off	Continuous kick-off and dead-time evidence: goal recall 45% to 74%. **Improved**
Nearest player is often not the actor	When our shooter is wrong, the true shooter is typically (median) 10.8 m from the tracked ball	Learned candidate ranking; blockers improved, shooter barely. **Data limit**
Headers and blocked shots missed	Found: headers 39% (foot shots 63%); blocked shots 42%	Trajectory context features gave a small gain. **Open**
Crosses and clearances look like shots	Most false alarms	PFF labels intention; movement is the same. **Open**
Shoot-outs not tracked	58 shots after the tracking video ends	Cannot be detected; without them recall is about 63%. **Reported**
Detection stuck at F1 0.65	8 times more data, a deeper model, context features and sequence stacking each changed F1 by less than 0.01	Limit of broadcast tracking; video features are the next step. **Data limit**

Limits to state openly. The unseen matches were scored twice (first and second model version); all design choices used training matches only. Not produced: deflector and inadvertent shots (too subtle or too rare), and shotType, keeperTouchType, badParry, saveable and grades (need video or human judgement).